03

CHANGELOOP

Question 3

What already exists, what is genuinely new here, and the difference between them.

The question

What is new here? What is the closest existing method, and how is yours different?

My answer

Let me name what already exists first, because most of this problem is served by technology that works.

Production planning and ERP systems already schedule dye lots against delivery dates. Counter-current rinsing already cuts water use inside the dyehouse, and has been Best Available Technique for decades. Low-electrolyte reactive dyes are a commercial product anyone can buy. Reverse osmosis, evaporators and ZLD control systems already handle the effluent once it arrives at the plant.

None of that is mine and I claim none of it. Running lots light to dark is ordinary dyehouse practice. If somebody on the panel says resequencing production is not a new idea, they are completely right.

What is new is where the decision happens, and what information is inside it when it does.

At the moment the chain runs one way. Production picks a plan, the effluent turns up at the treatment plant, and the cost appears afterwards on a different account. ChangeLoop moves that downstream cost into the upstream decision, so the plan is chosen with the consequence already visible.

Three things come out of that, and they are what I would actually defend.

Salt becomes the lever instead of water. Everything in this space counts litres, but in a closed loop the reject you have to boil is set by salt mass over the concentration ceiling. Water volume on its own barely touches the energy bill. The engine demonstrates it live: cut water 20 percent and evaporator energy moves 0.0 percent, cut salt 20 percent and it drops 20. A planner optimising water alone is pulling a lever that is not connected to anything.

The treatment cost is computed while planning rather than billed afterwards. Steam, carbon and treatment charge are evaluated for every candidate plan before one is picked.

And operational commitments are constraints, not penalties. A firm ship date cannot be traded away for an environmental gain of any size. In the worked example the plan that saves nearly half the fresh water gets refused, because it delays one order by 2.6 hours.

> A water tool that makes a factory miss a shipment gets switched off in a week. That refusal is the design working.

There is a fourth difference that matters commercially. This is advisory, not a controller. A person approves the plan and the quality gate stays human. I am not trying to replace ERP, MES, SCADA or ZLD control. I

am trying to connect two decisions that currently happen in separate rooms: what production picks today, and what treatment pays for tonight.

Supporting notes

Why the answer is shaped this way

The quickest way to lose a technical panel is claiming novelty over something they know is forty years old. Counter-current rinsing and low-salt chemistry are both well established, and a textile or chemical engineer will know it immediately. So everything gets conceded by name in the first three sentences, before anything is claimed.

Clearing away what is not new also does real work. Once that ground is given up, the remaining claim is narrow enough to defend properly. Integration and reframing are legitimate forms of innovation, but only if they are stated as that rather than dressed up as invention.

Evidence behind each claim

Claim	Basis
Counter-current rinsing is established BAT	EU IPPC BAT Reference Document for the Textiles Industry
Low-electrolyte reactive dyes are commercial	Reactive dyeing literature: 5–40 g/L low-salt against 50–80 g/L conventional
Ascending shade sequencing is standard practice	Stated as not-invented in `core/optimizer.py` and in the interface
0.0% against −20.0%	`zld.sensitivity_salt_vs_water()`, computed live
Option C refused at 2.6 h late	`figures.json` → `options.OPTION_C`, `firm_date_breaches = 1`
Constraints are feasibility, not penalty	`core/optimizer.py` → `check_constraints()`; an infeasible plan cannot be recommended at any objective value
Advisory, never autonomous	`automatic_release` is hardcoded `False` and no code path sets it true

What the answer deliberately avoids claiming

- No ownership of sequencing, rinsing methods, low-salt chemistry, RO, ZLD or evaporator technology.
- No suggestion that existing ERP or ZLD systems are bad or missing something. They solve their own problems well. The gap is between them.
- No claim to replace any of those systems.

The hardest follow-up

"So it's scheduling software with extra reporting?"

The scheduling is how it gets delivered, not what the contribution is. The contribution is that the objective being optimised now contains a consequence nobody prices at decision time, and that this consequence

turns out to be driven by a different variable than the whole industry is targeting. If it were really just "reorder your lots", the engine would not refuse its own lowest-water plan, and the 0.0 percent result would not exist to be found.